EXPERIMENT - 6

Snort IDS/IPS Using TryHackMe

Step 1: Login to Platform

Open TryHackMe in a browser and log in using your account.

Step 2: Open Snort Room

Search for a Snort-related room and join it to begin the experiment.

Step 3: Start AttackBox

Launch the AttackBox, which provides a ready-to-use Linux environment for performing the tasks.

Step 4: Verify Snort Installation

Open the terminal and confirm that Snort is installed and ready to use.

Step 5: Run Snort in IDS Mode

Start Snort in detection mode so that it monitors network traffic and generates alerts for suspicious activity.

Step 6: Detect Network Threats

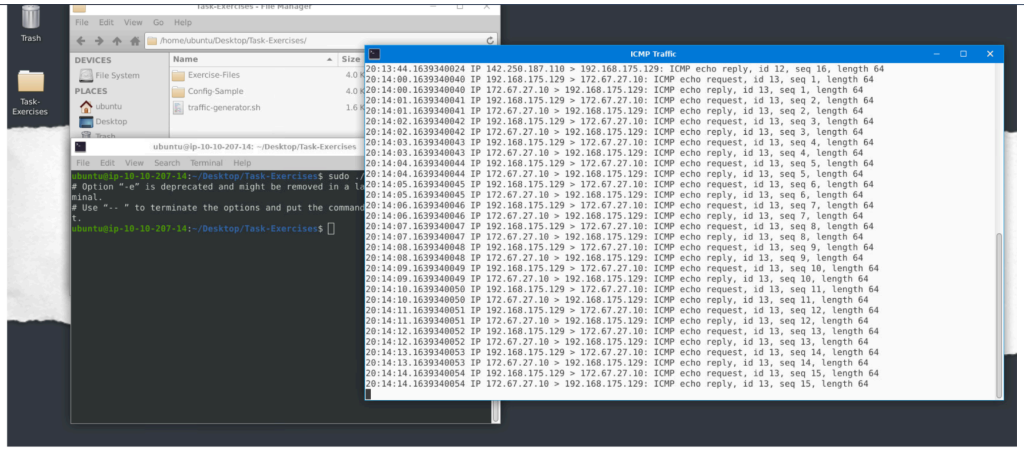
Perform the tasks given in the room and observe how Snort identifies and reports potential threats in real time.

Step 7: Configure IPS Mode

Enable prevention mode to allow Snort to block malicious traffic based on predefined rules.

Step 8: Analyze Alerts

Review the alerts generated by Snort and identify different types of network attacks.



Answer the questions below

Navigate to the Task-Exercises folder and run the command "./.easy.sh" and write the output

Too Easy!

✓ Correct Answer

Task 3 Introduction to IDS/IPS

Task 4 First Interaction with Snort

Once we use a configuration file, Snort has much more power! The configuration file is Snort's all-in-one management file. Rules, plugins, detection mechanisms, default actions, and output settings are identified here. Multiple configuration files are possible for different purposes and cases, but we can only use one at runtime.

Note that Snort automatically shows the default banner and initial information about your setup when you start it. You can prevent this by using the " **-q"** parameter.

Parameter	Description
-V / --version	This parameter provides information about your instance version.
-c	Identifying the configuration file
-T	Snort's self-test parameter allows you to test your setup with this parameter.
- **q**	Quiet mode prevents Snort from displaying the default banner and initial information about your setup.

That was easy; let's move on to the next task and continue exploring snort modes.

Answer the questions below

Run the Snort instance and check the build number.

149

✓ Correct Answer

Test the current instance with "/etc/snort/snort.conf" file and check how many rules are loaded with the current build.

4151

✓ Correct Answer

Test the current instance with "/etc/snort/snortv2.conf" file and check how many rules are loaded with the current build.

1

✓ Correct Answer

Identifying interfaces, note that Snort IPS requires at least two interfaces to work. Now run the traffic-generator script **as sudo** and start **ICMP/HTTP traffic**.

running IPS mode

```
user@ubuntu$ sudo snort -c /etc/snort/snort.conf -q -Q --daq afpacket -i eth0:eth1 -A console
Running in IPS mode

12/18-07:40:01.527100  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.175.131 -> 192.168.1
12/18-07:40:01.552811  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 172.217.169.142 -> 192.168.1
12/18-07:40:01.566232  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.175.131 -> 192.168.1
12/18-07:40:02.517903  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.1.18 -> 172.217.169.
12/18-07:40:02.550844  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 172.217.169.142 -> 192.168.1
^C*** Caught Int-Signal
```

As you can see in the picture above, Snort blocked the packets this time. **We used the same rule with a different action (drop/reject).** Remember, for the scope of this task, our point is the operating mode, not the rule.

Answer the questions below

Investigate the traffic with the default configuration file.

`sudo snort -c /etc/snort/snort.conf -A full -l .`

Execute the traffic generator script and choose **"TASK-7 Exercise"**. Wait until the traffic stops, then stop the Snort instance. Now analyse the output summary and answer the question.

`sudo ./traffic-generator.sh`

What is the number of the detected HTTP GET methods?

2 — ✓ Correct Answer

You can practice the rest of the parameters by using the traffic-generator script.

No answer needed — ✓ Correct Answer

Keep reading the output. How many TCP Segments are Queued?

18 — ✓ Correct Answer

Keep reading the output.How many "HTTP response headers" were extracted?

3 — ✓ Correct Answer

Investigate the **mx-1.pcap** file **with the second** configuration file.

`sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap`

What is the number of the generated alerts?

68 — ✓ Correct Answer

Investigate the **mx-2.pcap** file with the default configuration file.

`sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-2.pcap`

What is the number of the generated alerts?

340 — ✓ Correct Answer

Keep reading the output. What is the number of the detected TCP packets?

82 — ✓ Correct Answer

Investigate the **mx-2.pcap and mx-3.pcap** files with the default configuration file.

`sudo snort -c /etc/snort/snort.conf -A full -l . --pcap-list="mx-2.pcap mx-3.pcap"`

What is the number of the generated alerts?

1020 — ✓ Correct Answer

Result

Snort was successfully used as an IDS/IPS to detect and prevent network threats, demonstrating real-time network security monitoring in a controlled environment.